

Project Problem Proposal and Approval Form

Complete for Laboratory 2 and the ungraded Moodle approval checkpoint

Field	Team response
Team number	16
Working project title	Tsala Market
Members and student IDs	Thero Setlhare 201804745, Thusang Radingaka 202002221, Aobakwe Motingwa 201803690, Rodwell Patrick 201901812, Bonnie George 201801670
Repository URL	https://github.com/moffatrad/tsalamarket-CSI473
Proposal tag/commit	
Date submitted	10/08/2026

Originality declaration: This proposal is based on our own investigated problem. It is not the University Service Hub lecturer case.

1. Problem statement

Smallholder horticulture farmers across Botswana consistently sell their produce without knowing what it is actually worth. Unlike grain and pulse producers, who can at least reference the monthly producer prices published by the Botswana Agricultural Marketing Board (BAMB), horticulture growers have no equivalent pricing board at all — BAMB's mandate does not extend to fruits and vegetables, so no official body publishes a reference price for these crops. As a result, farm-gate prices for horticulture produce are set entirely by middlemen, disconnected from actual demand in urban markets like Gaborone and Francistown. The consequence is that horticulture farmers routinely accept below-market offers because they have no reliable reference point to negotiate from, eroding already-thin household incomes and reinforcing dependency on the buyers who benefit from that information gap. Without a way to close this asymmetry, horticulture smallholders remain structurally disadvantaged relative to buyers who always know the going rate, and the exclusion these farmers experience — falling entirely outside the price transparency that BAMB provides to grain producers — will persist.

2. Evidence of the problem

Evidence/source	Date/origin	What it shows
BAMB's monthly producer pricing structure and 11-branch network	BAMB operations, ongoing (current)	Confirms BAMB sets official prices for grains/pulses through a structured branch network, but its mandate does not extend to horticulture — illustrating the model of price transparency that horticulture producers currently lack
Absence of any BAMB-equivalent pricing board for horticulture produce	Market structure observation (current)	Confirms that fruits and vegetables fall entirely outside BAMB's price-publication mandate, leaving horticulture farmers with no official reference price and no equivalent of BAMB's branch network to consult
BAMB's bulk purchase/minimum-volume requirement (~200-bag threshold)	BAMB contract/purchase terms	Demonstrates a concrete, quantifiable barrier: most individual smallholders cannot produce enough volume to qualify for direct BAMB sales, pushing them toward informal buyers

3. Goal and objectives

Item	Team response
Goal	To reduce information asymmetry between smallholder horticulture farmers and buyers in Botswana by giving farmers accessible, verified price information — modelled on the transparency BAMB provides for grains — and a way to check whether an offer they've received is fair before they sell.
Objective 1	Design and implement a multi-channel price-lookup system (web dashboard and USSD/SMS) covering horticulture produce, sourced from major Botswana trading points, so farmers without smartphones or data can still check current prices — extending BAMB-style price transparency to a sector BAMB does not currently serve.
Objective 2	Develop a "Fair Price Check" feature where a farmer submits a price they've been offered, and the system compares it against the verified market average for that commodity and district, returning a clear fairness flag (fair / below average / significantly underpaid).
Objective 3	Implement a "Verified" vs "Community-reported" price-tagging and submission workflow — drawing on market vendors, cooperatives, and BAMB data — to build a crowd-sourced pricing dataset that is trustworthy enough to underpin the Fair Price Check calculation.

4. Stakeholders

Stakeholder/role	Need	Concern	Evidence/access
Smallholder Horticulture Farmer	Access accurate, timely prices before selling, and a way to check if an offer is fair	Reliability of the data source, and whether the system is usable without a smartphone or reliable data	Direct observation of rural farmer selling behaviour; USSD is the primary access point given low smartphone penetration
Market Vendor / Cooperative Lead	Coordinate group sales and submit or view price data on behalf of a ward/district	Time burden of manual data entry, and unclear incentive to participate consistently	Existing horticulture cooperative structures already used informally to coordinate group sales to urban markets
Verified Buyer (registered market trader / vendor)	Reach organised supply and post demand/pricing to relevant districts	Integrity of crowd-sourced price data feeding into the averages they're compared against	Existing produce-market vendor associations and municipal fresh-produce market records
Extension Officer / Ministry of Agriculture	Regional oversight of price trends and underpayment incidents to inform policy	Data privacy/anonymisation of farmer-level data, and staying within their advisory mandate	Existing district agricultural office reporting channels
System Admin	Manage user accounts, verify data sources, and maintain overall data integrity	Risk of fraudulent or manipulated price submissions undermining trust in the system	Admin logs and the verification workflow built into price submission

5. Scope, assumptions and constraints

In scope	Out of scope
Price lookup and historical trend viewing (web + USSD/SMS) for horticulture produce	Full ML-based price forecasting (only a simple trend/seasonal-average model, if attempted at all)
The "Fair Price Check" offer validator comparing a farmer's offer against the verified market average	Real payment processing or financial transactions between buyers and farmers
Verified vs. community-reported price tagging and submission workflow	Buyer trust/reputation rating system
Core user roles: Farmer, Vendor/Cooperative Lead, Verified Buyer, Extension Officer/Admin	Live integration with BAMB systems or real telecom USSD gateways (data will be synthetic/mock)

Assumptions	Constraints
Synthetic/anonymised price and user data will stand in for real horticulture farmer data during development and testing, since no BAMB-equivalent dataset exists for this sector	The semester timeline requires a feasible vertical slice rather than the full multi-tier feature set from the original concept
Users have access to at least basic mobile phones (USSD/SMS), even without smartphones or mobile data	No live integration with BAMB or telecom USSD gateways is possible — these will be simulated/mock

6. Proposed core workflow / vertical slice

[In 5-8 steps, describe one primary actor completing a useful task, the important rule to enforce, the state/result created, and one failure/exception to test.]

Primary actor: Smallholder Horticulture Farmer, completing a Fair Price Check on an offer they've just received.

1. The farmer dials the USSD short-code (or opens the app) and selects "Check My Offer."
2. The farmer selects their crop type and district, then enters the price a buyer has just offered them.
3. The system retrieves the current verified market average for that commodity and district from the price database.
4. **Important rule enforced:** only prices tagged "Verified" and submitted within the last 30 days are included in the average calculation — stale or unverified community-reported entries are excluded, so a single bad-faith submission can't skew the reference price (full normalisation, verification, aggregation and confidence methodology in Section 6a).
5. The system calculates the percentage deviation between the farmer's offered price and the verified average.
6. **State/result created:** the system returns a fairness flag — Fair, Below Average, or Significantly Underpaid — along with the reference average it used, and logs the (anonymised) query for the Extension Officer's aggregate dashboard.
7. The farmer decides whether to accept the offer, negotiate, or seek the nearest alternative buyer, based on the flag received.
8. **Failure/exception to test:** if no verified price exists for that commodity/district combination (e.g., an under-reported area or an uncommon crop), the system does not fail silently or guess — it tells the farmer insufficient data is available and offers the nearest district or a broader regional average as a fallback instead.

6a. Price normalisation, verification, aggregation and classification methodology

Horticulture produce is not covered by BAMB's grading and pricing system, so this methodology is defined by the project rather than sourced from an official pricing board. It is modelled on the structure BAMB uses for grains and pulses — standard units, graded tiers, and a published validity window — but adapted for informally traded, largely ungraded horticulture produce.

Units: All prices are normalised to Pula per kilogram (P/kg) before they are stored, averaged, or displayed. Submissions given in local sale units (per crate, per 50kg-equivalent bag, per heap/"tshelo", per bundle) are converted at entry using a commodity-specific conversion-factor table maintained by System Admins (e.g., a standard tomato crate $\approx$ 20kg); a submission in an unrecognised unit is rejected at entry rather than guessed or silently dropped.

Grade / quality tier: In place of an official grade, the farmer or vendor self-declares one of three quality tiers at submission — Grade A (export/premium), Grade B (standard market quality), or Grade C (below-standard/surplus) — mirroring BAMB's Grade 1–3 structure shown on its producer price list. An entry submitted with no declared grade defaults to Grade B for the purpose of the average (a conservative, middle assumption) and is flagged "ungraded" in the Admin view so it can be corrected.

Market level: Every price is tagged Farm-gate, Wholesale/municipal market, or Retail, since the same commodity trades at very different prices at each level. The Fair Price Check always compares a farmer's offer against the Farm-gate verified average for the matching commodity, district and grade — never against wholesale or retail prices — so the comparison stays like-for-like.

Verification: An entry is promoted from "Community-reported" to "Verified" only if: (a) it is submitted by a registered Verified Buyer, Market Vendor or Cooperative Lead account; (b) it is corroborated by a second independent submission for the same commodity/district/grade/market-level within 72 hours and within 15% of the first; or (c) a System Admin confirms it manually. Unverified entries stay visible for trend context but are excluded from the Fair Price Check calculation, extending Objective 3's verification workflow.

Aggregation window and minimum sample size: Only Verified entries from the trailing 30 days are pooled, bucketed by commodity $\times$ district $\times$ grade $\times$ market level. A bucket must contain at least 5 independent Verified entries, from at least 2 different submitters, before it is used to generate a Fair Price Check result; below that threshold the system applies the existing fallback rule (nearest district or a broader regional bucket) and marks the result "Low Data" rather than returning a thin, unreliable average.

Outlier handling: Before an average is calculated, entries more than $1.5 \times$ the interquartile range (IQR) outside the bucket's Q1–Q3 range are treated as outliers, excluded from the calculation, and logged for Admin review rather than silently discarded — so a single mistyped or opportunistic quote cannot swing a thin dataset.

Central tendency: The reference price used for comparison is the median (not the mean) of the remaining pooled entries. The median is less sensitive than the mean to the small, unevenly distributed samples expected in an early-stage crowd-sourced dataset.

Confidence rating: Each Fair Price Check result carries a confidence label derived from sample size, submitter diversity and recency: High (≥ 10 Verified entries from ≥ 3 submitters, all within 7 days), Medium (5–9 entries, or data 8–30 days old), or Low (a fallback to a neighbouring district or regional bucket). The confidence label is shown to the farmer alongside the fairness flag, not hidden in the background.

Classification thresholds: The fairness flag is set by the offered price's percentage deviation below the reference median: within $\pm 10\%$ = Fair; 10–25% below = Below Average; more than 25% below = Significantly Underpaid. These thresholds, together with the 30-day window, 72-hour corroboration limit and 5-entry minimum above, are Admin-configurable parameters rather than fixed assumptions, so they can be recalibrated as real data accumulates.

Presented as indicative guidance, not a guarantee: Every Fair Price Check result is shown with an explicit statement that it is indicative guidance based on recent aggregated verified submissions, not a guaranteed, audited or binding price. It does not account for produce quality variation, transport cost, or buyer-specific circumstances, and the farmer is directed to still weigh those before deciding whether to accept, negotiate, or seek an alternative buyer.

7. Design-suitability check

Check	Yes/No	Short evidence
Clear stakeholders and a genuine current problem	Yes	Farmers, vendors/cooperatives, verified buyers, and Ministry extension officers are all identified; the problem is grounded in BAMB's documented structure for grains and the total absence of an equivalent pricing mechanism for horticulture
At least one state-sensitive workflow or entity	Yes	Price entries move through unverified → verified states; a farmer's offer query moves through submitted → evaluated states in the Fair Price Check
Important business/validation/security rule	Yes	Only "Verified" price entries submitted within the last 30 days feed the fairness-check average — stale or unverified data is excluded from the calculation
Relevant quality trade-offs	Yes	Trade-off between data freshness/coverage (allowing more crowd-sourced input) and data reliability (requiring a verification step before data is trusted)
Relevant failure/exception condition	Yes	No verified price exists for a given commodity/district combination, requiring a fallback response rather than a silent or misleading result
Feasible semester-sized vertical slice	Yes	Scope is limited to price lookup plus the Fair Price Check for a defined subset of commodities/districts, excluding forecasting, payments, and reputation features
Synthetic/anonymised data is sufficient	Yes	Historical horticulture market prices and averages will be synthesised/mockd rather than sourced from live BAMB or telecom systems, since BAMB does not publish horticulture prices
Not the University Service Hub or a renamed copy	Yes	The problem is an original agricultural market-transparency system for Botswana smallholders, unrelated in domain and structure to the University Service Hub case

8. Data, ethics, safety and dependency risks

Risk/concern	How the team will control it
Farmer/user data privacy (location, phone number, price submissions)	Store location data at district level only rather than precise coordinates; collect no personally identifiable information beyond what login requires
Reliance on synthetic/mockd horticulture market price data rather than live feeds, given no BAMB or equivalent live source exists	Clearly document all price data as synthetic/for demonstration purposes; design the data model so real data sources can be substituted later without changing the architecture
Risk of inaccurate or malicious crowd-sourced price submissions	Enforce the "Verified" vs "Community-reported" tagging system, and restrict the Fair Price Check calculation to verified, recent entries only

9. Lecturer decision

Decision	Select one
Approved	☐
Approved with minor conditions	☐
Revise and resubmit	☐

Field	Complete
Conditions/comments	
Lecturer	Dr Cleverence Kombe
Decision date / resubmission date if applicable	